

Q-Learning vs. SARSA for ICU Sepsis Treatment: A Tabular Reinforcement Learning Study with DQN Extension

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1. Executive Summary

This project studies ICU sepsis treatment as a sequential decision-making problem using the published ICU-Sepsis benchmark, where an agent observes a discrete patient state and selects a treatment action combining intravenous fluids and vasopressors. We implemented and compared two tabular model-free TD control methods, Q-learning and SARSA, and added DQN as an advanced extension. The learned tabular policies select state-dependent treatment actions by choosing the highest-value action from the final Q-table. In the final 300,000-episode experiment, Q-learning improved fastest early but showed higher variability, ending at 0.7856 ± 0.0009 , while SARSA learned more gradually and achieved the strongest tabular result, 0.7880 ± 0.0007 . DQN achieved the highest overall return as an extension, 0.7913 ± 0.0035 , but its improvement over SARSA was small relative to its added complexity. Overall, the agents learned policies near or above the random baseline, but the gains remained modest relative to the benchmark optimal return of approximately 0.88.

2. Problem Formulation and Final MDP

We use the published ICU-Sepsis benchmark [1], built from records of approximately 17,000 sepsis patients in MIMIC-III, to model ICU sepsis treatment as a finite, discrete MDP. This allows us to focus on RL methodology within a standardized tabular setting rather than reconstructing the environment from raw clinical data.

The state space contains 716 discrete patient states, each representing a K-means cluster of 47 normalized clinical features measured in 4-hour blocks. The action space contains 25 treatment actions, arranged as a 5×5 grid over intravenous-fluid and vasopressor dosage levels. The reward is sparse and terminal: intermediate rewards are 0, survival gives +1, and death gives 0. With $\gamma = 1$, expected return represents simulated survival probability.

3. Algorithmic Methodology

We implemented two tabular model-free TD control methods: Q-learning and SARSA. Both are suitable because the ICU-Sepsis benchmark has finite discrete states and actions, allowing the action-value function to be stored as a 716×25 Q-table. We selected these methods to compare off-policy and on-policy learning under the same training pipeline. Q-learning uses the target $r + \gamma \max_a Q(s', a')$, updating toward the best estimated next action, while SARSA uses $r + \gamma Q(s', a')$, updating toward the next action actually selected by the current behavior policy.

Both agents used ϵ -greedy exploration. At each step, the agent selected a random action with probability ϵ and otherwise selected the action with the highest current Q-value. ϵ decayed from 1.0 to 0.001, shifting training from broad early exploration toward later exploitation of the learned policy.

Hyperparameters were selected using a simplified grid search during the tuning stage. We tested learning rates $\alpha \in \{0.001, 0.0025, 0.01\}$, exploration fractions $\{0.10, 0.25\}$, and seeds $\{5, 15, 25\}$, with 500,000 episodes per tuning configuration. We used $\gamma = 1.0$, following the benchmark, because undiscounted return corresponds to simulated survival probability. The final settings were $\alpha = 0.01$ and exploration fraction 0.10 for Q-learning, and $\alpha = 0.01$ and exploration fraction 0.25 for SARSA, chosen based on final average return across seeds. Compared with the benchmark paper, our exploration fractions were consistent, while the higher learning rate may reflect faster convergence in our smaller tuning setup; we therefore treat these settings as coarse empirical choices rather than fully robust optima.

4. Experimental Results and Performance Evaluation

Figure 1 compares Q-learning, SARSA, and DQN over 300,000 training episodes, with shaded regions showing standard error across seeds. The tabular curves are averaged over 10 seeds, while DQN is averaged over 3 seeds. The figure also includes two benchmark references: a random-policy baseline of approximately 0.78 and an optimal value-iteration upper bound of approximately 0.88. These references allow us to assess whether the agents learned policies better than random treatment selection and how far they remain from the benchmark optimum.

The learning curves show distinct behavior. Q-learning improved fastest in the early stage but later plateaued with higher seed variability, ending at 0.7856 ± 0.0009 . SARSA improved more gradually but produced the strongest and most stable tabular result, reaching 0.7880 ± 0.0007 . DQN started more slowly but continued improving and achieved the highest final average return, 0.7913 ± 0.0035 . Overall, all methods performed near or above the random baseline, suggesting that the agents learned useful policies rather than behaving like uniformly random action selectors. However, the improvements were modest, and none of the methods closed most of the gap to the 0.88 optimal benchmark return.

5. Analysis and Interpretation

The learned tabular policies are represented by the greedy actions selected from the final Q-table for each of the 716 states. Because the states are anonymized K-means clusters rather than directly interpretable clinical categories, we interpret these policies only at the benchmark level, not as clinical treatment rules or medical recommendations.

The distinct learning curves are consistent with the algorithms' update rules. Q-learning's faster early improvement reflects its off-policy target, which can propagate high values rapidly but may

also amplify overestimation when Q-value estimates are noisy. SARSA's slower but more stable trajectory reflects its on-policy update, which accounts for the exploratory randomness still present during training.

The main limitations are the smaller-scale reproduction protocol, limited hyperparameter search, sparse terminal reward, and restricted policy interpretability. The benchmark paper tuned hyperparameters using 8 random seeds per setting and reported final learning curves over 1,000 random seeds, whereas our final tabular experiments used 10 seeds and our DQN extension used 3 seeds because of runtime constraints. Therefore, our results should be interpreted as course-scale reproduction estimates rather than fully robust replications of the paper's performance. In addition, our tuning used a small grid of learning rates and exploration fractions, so the selected hyperparameters remain coarse empirical choices. The terminal-only survival reward makes credit assignment difficult, and the 716 K-means states compress 47 clinical features, limiting direct clinical interpretation of the learned policies.

6. Bonus: Advanced Extensions

As an advanced extension, we implemented DQN by replacing the Q-table with a neural-network Q-function. The state ID was encoded as a 716-dimensional one-hot vector and passed through a two-layer MLP with 128 hidden units to output Q-values for 25 actions. DQN used a target network, replay buffer, mini-batch updates, and ϵ -greedy exploration. It was trained for 300,000 episodes across 3 seeds; we used fewer seeds than the tabular methods because DQN training was substantially slower due to neural-network optimization, replay sampling, and target-network updates.

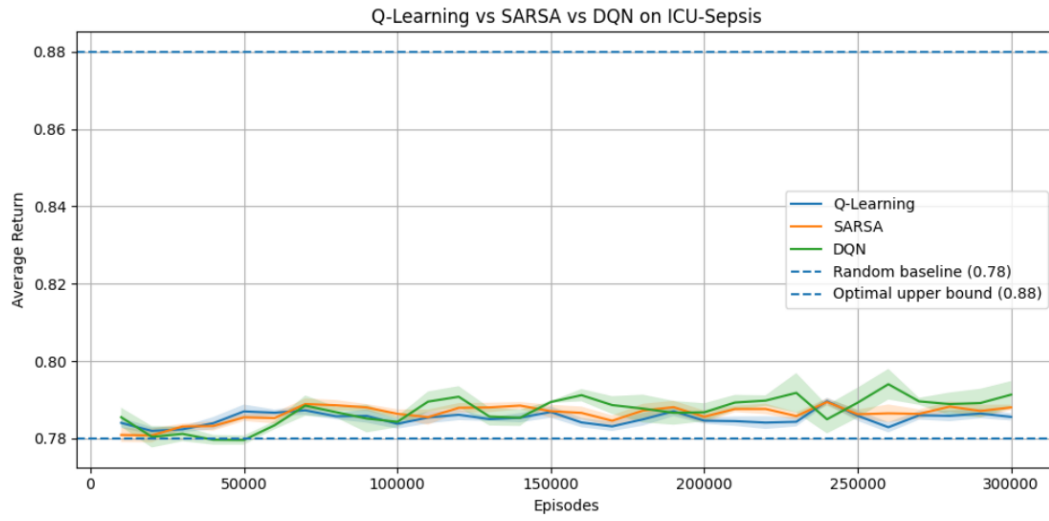
As reported in Section 4, DQN achieved the highest final return. The trade-off is that DQN is more scalable and can generalize across states, but its improvement over the tabular methods was small relative to its added complexity. In this 716-state benchmark, the tabular methods already represent every state-action value directly, so DQN's function approximation provided limited practical gain despite requiring more computation.

References

[1] Choudhary, K., Gupta, D., & Thomas, P. S. (2024). ICU-Sepsis: A Benchmark MDP Built from Real Medical Data. *Reinforcement Learning Journal*, 4, 1546–1566.

Appendix

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Final results averaged over seeds:
Q-Learning : 0.7856 ± 0.0009 (paper: ~ 0.84)
SARSA : 0.7880 ± 0.0007 (paper: ~ 0.79)
DQN : 0.7913 ± 0.0035 (paper: ~ 0.86)

Figure 1: Learning curves for Q-learning, SARSA, and DQN on ICU-Sepsis